

Mountains

In many land habitats, climatic conditions vary only slightly within a region. In mountains, however, the average air temperature drops by about 1.8° F (1° C) for every 650 ft (200 m) gained in height, oxygen becomes more scarce, and the air becomes less effective at screening out ultraviolet light. As a result, mountains can be divided into distinct zones, each supporting plant and animal life that is very different from that of the zone above and below. A wide range of animals live in the low-elevation foothills, but only the hardest survive year-round in the harsh environment above the tree line.

Temperate mountains

In temperate regions, a mountain's climate is relatively cool throughout the year. However, seasonal changes are much more marked than they are in the tropics. At high elevation, above the tree line, there is a sudden burst of plant growth in spring and summer. Some animals migrate upward to make use of this brief abundance of food, but others, such as the marmot, are permanent residents between mid and high elevation, surviving the winter cold by living in burrows, and by hibernating for up to 8 months a year. High-elevation insects spend many months in a dormant state, coming to life when warm weather arrives. For many, this

dormant period is spent inside the egg, which hatches when the days lengthen and the temperatures rise.

At lower elevations, the climate is warmer, and generally more like that of the surrounding land. However, because the sloping, rocky ground is difficult to farm, the mountainsides often

retain more of their tree cover than does flatter ground. In undisturbed conditions, these montane forests are the natural habitat of large mammals, such as pumas, bears, and deer, and also of a wide range of seed- and insect-eating birds.

Temperate mountains abound in birds of prey. Some, such as the peregrine falcon, pursue their prey on fast-flapping wings, while eagles and buzzards soar high up, riding on updrafts. One characteristic mountain species, a vulture called the lammergeier, turns the mountain landscape to its advantage by carrying carrion bones aloft and dropping them onto the rocks to expose the edible marrow inside.



THE SNOW LEOPARD spends its summers above the treeline, but during winter follows hoofed animal prey to lower elevations.



THE EURASIAN BLACK VULTURE is wide-ranging across open country, scavenging on carrion, but occasionally taking live prey, such as rabbits.



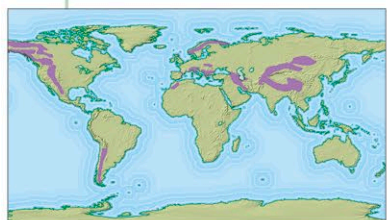
BIGHORN SHEEP live on rugged terrain, from precipitous slopes of cliffs and rocky outcrops down to hot canyons and desert valleys.



ALPINE MARMOTS can spend half the year in hibernation—one of the longest periods of dormancy for any mammal.



A DIVERSITY OF mountain frogs thrives around streams in moist, humid foothill forests—with many species restricted to isolated ranges.



THE NORTHERN temperate zone's mountain ranges include the highest peaks on earth. In the Southern Hemisphere, mountains are smaller and more isolated.

HIGH ELEVATION

MID-ELEVATION

LOW ELEVATION

